

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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quartile serves as the reference category. For firm-level job postings, effects are significant across all higher quartiles, though the third and fourth quartiles show similar magnitudes, which is consistent with a threshold effect. For firm-level employment stocks, the evidence is weaker and non-monotonic, with only scattered significant effects across bins. The occupation-level analysis reveals clearer nonlinearities: while the second and third quartiles show minimal effects, the highest exposure quartile experiences significantly larger impacts, suggesting that most of the occupation-level results we observe are driven by the 25 percent most exposed occupations.

5 Conclusion

This paper provides evidence that large language models are reshaping UK labor markets. Using a difference-in-differences design that exploits variation in occupational- and firm-level LLM exposure, we conduct three analyses that show how labor markets change in response to the introduction of generative AI. First, we examine firm-level employment stocks to understand how workforce size and composition evolve within organizations. Second, we analyze firm-level hiring patterns to capture forward-looking recruitment decisions that precede employment adjustments. Third, we investigate occupation-level posting dynamics to identify economy-wide demand shifts.

Our firm-level employment analysis documents a 0.3 percent reduction in total employment per standard deviation increase in LLM exposure, though this effect only becomes statistically significant approximately 18 months after ChatGPT’s introduction. This reduction is predominantly concentrated in junior positions, which decline by 0.4 percent per standard deviation, while senior positions show a smaller, statistically insignificant 0.2 percent reduction. The resulting compositional shift reduces the fraction of junior workers by 0.1 percentage point per standard deviation, mechanically increasing average compensation by £106 as firms retain proportionally more higher-paid senior staff, though this compensation effect is only marginally significant.

While employment stocks adjust gradually, job posting data provide real-time evidence of firms’ evolving labor demand. Firm-level posting activity drops following

ChatGPT’s release, with a one standard deviation increase in LLM exposure reducing the probability of posting any vacancy by 1.0 percentage point. This reduction emerges gradually and becomes significant after approximately 10 months, mirroring the timeline of employment stock adjustments. When we examine the composition of roles driving this reduction, we find that technical positions and creative roles experience the sharpest declines. Sales representative positions, by contrast, show a statistically significant increase, suggesting that customer-facing roles requiring direct interpersonal interaction remain robust to LLM-based automation.

In addition to within-firm dynamics, our occupation-level analysis reveals how labor demand shifts for entire job categories regardless of firm exposure. A one standard deviation increase in LLM exposure is associated with a 3.9 percent reduction in posting volumes, accompanied by £495 (1.1 percent) salary declines. The apparent inconsistency of declining posted salaries for exposed occupations alongside increasing firm-level compensation shows how aggregation could mask opposing forces. While LLM exposure drives down market wages for specific occupations, within-firm rebalancing toward more senior positions pushes average firm-level wages in the opposite direction.

Our heterogeneity analysis reveals that the observed labor market disruptions are almost entirely driven by high-remuneration segments of the labor market. This concentration of effects in higher-wage segments contradicts historical automation patterns that predominantly displaced routine middle-skill work, and suggests that LLMs might lead to a compression of the wage distribution.

Our findings carry important implications for understanding technological displacement and designing policy responses. The concentration of job losses among entry-level positions disrupts traditional skill development pathways where workers master progressively complex tasks through hands-on experience. Without junior roles serving as training grounds, firms may struggle to develop senior talent internally while new entrants face shrinking opportunities to gain experience (Ide, 2025). The freeze in hiring followed by employment stock changes suggests that forward-looking indicators like job postings may provide early warning signals for policymakers. The sharp reductions in technical and analytical roles alongside resilience in sales positions indicates that human interaction and relationship-building skills remain valuable complements to LLM capabilities, suggesting that retraining programs should emphasize these interpersonal competencies. Policy responses